

Youngmin Jeong

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Research Interests

Computer Vision, Robot Vision, Visual Perception, Multi-modal Learning, Robot Manipulation

Education

Seoul National University of Science and Technology (SeoulTech)

Seoul, Korea

B.S. in Applied Artificial Intelligence, GPA: 3.9/4.5, Expected Graduation: Feb. 2027

Mar. 2021 – Present

Research Experience

Intelligent Pattern Analysis and Learning (IPAL) Lab

Seoul, Korea

Undergraduate Researcher (Advisor: Prof. Beom-Seok Oh)

Oct. 2024 – Aug. 2026

Artificial Intelligence for Remote Sensing (AIR) Lab

Seoul, Korea

Undergraduate Researcher (Advisor: Prof. Seonyoung Park)

Apr. 2021 – Feb. 2023

Publications

⟨ **International Journal** ⟩ (*Equal contribution is denoted by “*”.*)

[1] Yerang Lee*, **Youngmin Jeong***, Beom-Seok Oh, Seonyoung Park, and Eun Gyung Chung, “A Deep Semantic Segmentation System for Detecting Multi-Scale Barns of Diverse Livestock Breeds in Satellite Imagery,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (J-STARS)*, 2026. (*Under Review*)

⟨ **Domestic Conferences** ⟩

[1] **Youngmin Jeong** and Beom-Seok Oh, “GLCFormer: Improving Livestock Facility Detection in Satellite Imagery Using Graph-Based Land-Cover Information Fusion,” *Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE)*, 2026.

[2] Yerang Lee*, **Youngmin Jeong***, and Beom-Seok Oh, “HeteroBarnMapper: A Detection-to-Mapping System for Heterogeneous Livestock Facilities with Reliable Evaluation,” *Summer Annual Conference of IEIE*, 2025. (*Oral Presentation*)

[3] **Youngmin Jeong***, Yerang Lee*, and Beom-Seok Oh, “DualLCFormer: Livestock Facility Detection using a Dual Encoder Architecture with Land Cover Fusion,” *Summer Annual Conference of IEIE*, 2025.

[4] **Youngmin Jeong**, Kyungil Lee, and Seonyoung Park, “Landsat 9 based Local Climate Zone Mapping using CNN in Rome,” *Fall Annual Conference of the Korean Association of Geographic Information Studies (KAGIS)*, 2022. (*Extended Abstract*)

Research Projects

Dataset Construction and Management for Training Satellite Imagery Foundation Models

Mar. 2026 – Present

Korean Society for Aeronautical and Space Sciences (KSAS)

- **Role:** Foundation model implementation and evaluation using curated multi-source satellite imagery datasets.

Development of Monitoring Methods for Livestock Facility Detection

Nov. 2024 – Nov. 2025

National Institute of Environmental Research (NIER)

- **Role:** Led model development and extensive experimentation for an end-to-end livestock facility detection pipeline, including post-processing, object-level evaluation, and field validation.

Selected Projects

RS-AutoLabel: Training-Free Open-Vocabulary Remote Sensing Auto-Labeling

Sep. 2025 – Jun. 2026

Capstone Design Project, SeoulTech

- Developed an open-vocabulary remote sensing auto-labeling system with region consistency refinement and user-assisted exemplar prompting.

Open-Vocabulary VLM-Based Interactive Object Labeling Automation System

Sep. 2025 – Nov. 2025

2025 Space Hackathon, Korea Aerospace Research Institute (KARI)

- Developed an interactive object labeling automation system using open-vocabulary vision-language models.

Awards & Honors

President’s Award, 8th National Undergraduate Capstone Design Competition, The Society for Aerospace System Engineering (SASE)

Aug. 2026

Special Award, 2026 Capstone Design Competition, SeoulTech

Jun. 2026

Top Excellence Award (KARI President’s Award), 2025 Space Hackathon

Nov. 2025

AIS Scholarship (Full Tuition), SeoulTech

Mar. 2021 – Present